UNIT 2

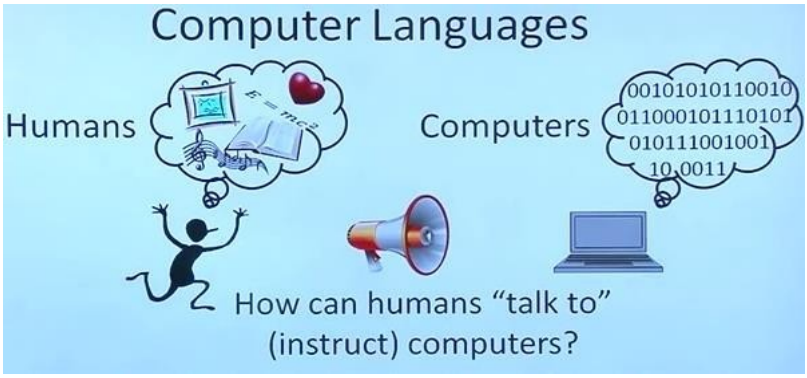
PROGRAMMING LANGUAGE



GLOSSARY

WORD	DEFINITION
Low level language	this code is written to specific hardware, and will only operate on the hardware it was written for and has almost no abstraction from the hardware; reads machine code and assembly language
APPLICATION	A program which makes the computer a useful tool.
Compiler	Converts a program to a language that the computer understands.
Scripts	In programming, a series of scripts, or sets of steps, are written for a computer to follow. Computers process the steps line-by-line from top to bottom. Each step is created by writing a statement.
Compile	It is to transform a program written in a high-level programming language from source code into object code. A compiler reads the whole source code and translates it into a complete machine code program to perform the required tasks which is output as a new file.
Code	In computer programming, computer code refers to the set of instructions, or a system of rules, written in a particular programming language
High-level language	A high-level language (HLL) is a programming language that lets the developer write programs irrespective of the nature or type of computer. If a computer has to understand a high-level language, it should be compiled into a machine language.
Low-level language	A low-level language is a language that is very close to machine language and provides a little abstraction of programming concepts. Low-level languages are closer to the hardware than human languages. The most common examples of low-level languages are assembly and machine code.
Markup language	A markup language is a relatively simple language that consists of easily understood keywords and tags, used to format the overall view of the page and its contents.
ASCII	Short for **American Standard Code for Information Interexchange**, **ASCII** is a standard that assigns letters, numbers, and other characters
HTML	First developed by Tim Berners-Lee in 1990, **HTML** is short for **Hypertext Markup Language**. HTML is used to create electronic documents (called pages) that are displayed on the World Wide Web. Each page contains several connections to other pages called hyperlinks.

1.1. WHAT IS A COMPUTER LANGUAGE?



A **programming language** is a computer language programmers used to develop software programs, scripts, or other sets of instructions for computers to execute.

Programming languages are leading the industry due to their versatility, robustness, and widespread adoption.A programming language is **a computer language engineered to create a standard form of commands**. These commands can be interpreted into a code understood by a machine.

Programming is an important engineering tool , It is a process of writing a computer program using computer language. Computer programs are collections of instructions that tell a computer hardware and how to process data. Our work would have been very demanding and time consuming without programming.

Each programming language has an unique set of keywords along with a special syntax to organize the software's instructions.

1.2. TYPES OF PROGRAMMING LANGUAGES

programming languages can be broken into two types of languages that are:

1.2.1. HIGH-LEVEL LANGUAGE (MOST COMMON)

High-level languages are more human-readable and abstracted from hardware, making them easier to program but less efficient.

High-level languages, on the other hand, are designed to be easy to read and understand, allowing programmers to write source code naturally, using logical words and symbols.

Sometimes abbreviated as HLL, a high-level language is a computer programming language that isn't limited by the computer, designed for a specific job, and is easier to understand. It is more like human language and less like machine language. However, for a computer to understand and run a program created with a high-level language, it must be compiled into machine language.

1.2.2. LOW-LEVEL LANGUAGE

Low-level languages are closer to machine code, offering more control and efficiency but are harder to program**.**

Low-level languages include assembly and machine languages. An assembly language contains a list of basic instructions and is much harder to read than a high-level language.

A low-level language is a programming language that provides little or no abstraction of programming concepts and is very close to writing actual machine instructions. Two examples of low-level languages are assembly and machine code.

Low-level languages are useful because programs written in them can be crafted to run very fast and with a very minimal memory footprint. However, they are considered harder to utilize because they require a deeper knowledge of machine language.

1.3. MAIN FEATURES OF PROGRAMMING LANGUAGES

The features that a programming language must have to stand out are the following:

1.3.1. **Simplicity:** the language must offer clear and simple concepts that facilitate learning and application, in a way that is simple to understand and maintain. Simplicity is a difficult balance to strike without compromise the overall capability.

2.3.2 .Naturalness: this means that its application in the area for which it was designed must be done naturally, providing operators, structures and syntax for operators to work efficiently.

2.3.3. Abstraction: it is the ability to define and use complicated structures or operations while ignoring certain low-level details.

2.3.4. Efficiency: Programming languages must be translated and executed efficiently so as not to consume too much memory or require too much time.

2.3.5. Structuring: the language allows programmers to write their code according to structured programming concepts, to avoid creating errors.

2.3.6. Compactness: with this characteristic, it is possible to express operations concisely, without having to write too many details.

2.3.7 Locality: refers to the code concentrating on the part of the program with which you are working at a given time.

2,4. THE MOST POPULAR PROGRAMMING LANGUAGES

2.4.1. PYTHON



Python remains a dominant force across industries, from web development to artificial intelligence and data analytics. Its simplicity and vast library ecosystem make it a top choice for developers worldwide.

Python has all the data science and machine learning libraries which makes it more relevant than ever in 2025 with so many companies madly rushing to adopt AI. Plus, it's an incredibly easy language to learn for beginners.

Python is a *high-level*, general-purpose programming language. It can be used for various tasks, from data analysis and visualization to web development, prototyping, and automation. Python is popular among software developers since it works well as a scripting language. Moreover, it lets users utilize *different programming paradigms*, including the object-oriented, imperative, procedural, and functional programming languages.

Python is **easier** to learn than other languages thanks to its **syntax** familiarity with English and to be popular **among** software **developers**. If you want to kickstart your career as a software developer or engineer, *learning to code* in Python can be a great starting point.

Python can be regarded as the future of programming languages. As per the latest statistics, **Python is the main coding language for around 80% of developers.** The presence of extensive libraries in Python facilitates artificial intelligence, data science, and machine learning processes. Currently, Python is trending and can be regarded as the king of programming languages

It is one of the most lucrative languages that offers amazing features like:

- Easy to learn and code.
- Extensive libraries and frameworks that support a plethora of applications.
- Incorporated the variants of Java and C like CPython, Jython, etc.
- GUI support.
- Companies working on Python: Intel, Facebook, Spotify, Netflix, etc.

Python Advantages and Disadvantages

Advantages	Disadvantages
• Easy to read and learn • Reduces maintenance costs • Easy Memory Management • Wide applicability • Large community • Easy integration with other programming languages • Asynchronous coding • Portable • Enterprise Application Integration	• Slow execution speed • Large memory consumption • Not ideal for mobile and game development • Addiction to easy coding • Weak database access • Design restrictions • More prone to runtime errors

2.4.2. **JAVA**



Java is an easy coding programming language owned by Oracle. It's a high-level, general-purpose programming language, enabling coders to create all sorts of applications easily.

On top of that, the new Java frameworks, make it more convenient for developers to build applications.

Java remains integral to industries such as finance, healthcare, and e-commerce and the Recent Update introduces advanced memory management, structured concurrency, and performance enhancements, further strengthening Java's enterprise capabilities.

Why is it so popular? Java can run on all types of devices, it's open source, has a wide community of developers, and it's also easy to learn and use.

Java is also very popular among enterprise-sized companies because Java applications are stable, efficient, and scalable—all things you want when having an application dealing with potentially millions of transactions. That makes it a worthy addition to any developer's resume.

Java is one of the **most powerful programming languages** that is currently used in more than **3 billion devices**. Java is currently one of the most trending technology. It is used in desktop applications, mobile applications, web development, Artificial intelligence, cloud applications, and many more.

Some of the prominent features of Java are:

- *Platform independence and Object-oriented programming*
- *Enhanced productivity, performance, and security*
- *It is the most secure language*

2.4.3. JAVASCRIPT



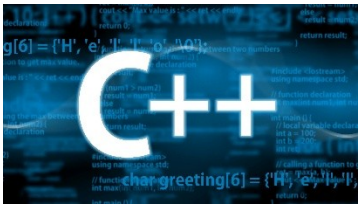
It's a vital tool for web developers and it is easy to learn as a high-level interpreted language.

This scripting language is often used to improve a webpage's interactivity. It enables web developers to add dynamic elements on their landing pages, such as animated graphics, clickable buttons, and mouseover effects.

JavaScript is also an excellent language for developing web apps as it offers an intuitive design.

- *It has a well-organized codebase that provides enhanced productivity and readability.*
- *Easy to learn and is highly in demand.*
- *Platform independence and greater control of the browser.*
- *Provide user input validation features.*
- *The top companies using JavaScript are Microsoft, Uber, PayPal, Google, Walmart, etc.*

2.4.4. C++



C++ is the enhanced version of C and one of the most popular programming languages in computer science. What's more, versatility makes C++ the best programming language to learn.

Its fast and powerful nature lets **developers create applications with excellent performance, like video games, graphics software, and web browsers.** It combines low and high-level language characteristics, making it excellent for system programming .

In terms of job opportunities, C++ proficiency is necessary for many IT professionals, such as software developers and programming architects. At present, C++ developer jobs are in high demand as it's one of the most complex coding languages to learn.

2.4.5. **C#**



C# is a key language for .NET development, known for enterprise solutions, game development, and fintech applications. Its versatility continues to make it a top choice across multiple industries. Recent Update: **C# 13** introduces features to streamline code syntax and improve performance making development faster and more efficient.

It is one of the top programming languages due to its fast performance and improved stability. This coding language is also easier to learn than its predecessors, C and C++, as it has a simple syntax and well-defined class hierarchy.

Starting off as Microsoft's language for its .NET framework, C# is now compatible with Windows and Linux. Therefore, it's arguably the best programming language to learn for developing GUI-based desktop applications.

What is C# used for? Like other general-purpose programming languages, C# can be used to **create a number of different programs and applications: mobile apps, desktop apps, cloud-**

based services, websites, enterprise software and games.

2.4.6. PHP



PHP is a widely-used, open-sourced, server-side language used for web development. it's open source and it is supported by an extensive community and body of resources, and it is easy to learn. On the other hand, many developers claim it is messily designed, and as a result leads to developers wh write bad PHP code.

2.4.7. GO

https://www.youtube.com/watch?v=CWZL0sE-2CE



Go, or Golang was created for developing APIs, GUI-based desktop applications, and web applications. While it's a young language, Go is one of the fastest-growing programming languages.

Concurrent programming is one of Go's best features. It enables programmers to work with multicore CPUs and a massive codebase.Combined with its excellent performance, Go concurrency makes it the perfect coding language to create algorithms, web servers, and data pipelines.

Go's concurrency is relatively simpler than other coding languages. With the lightweight Goroutines thread, programmers only need to add the go syntax to make a function run as a subprocess.

Some of the best job roles for Go developers include software engineers, data scientists, back-end web developers, and AI researchers.

2.4.8. SQL Structured Query Language



Structured Query Language (SQL) is the best programming language to learn if you're interested in data science and statistical computing. It's a domain-specific language that lets programmers query, manipulate, and analyze data stored in a relational database.

SQL is the *third most-used language* based on Stack Overflow's statistics. In terms of job opportunities, companies from all industries may need SQL developers as they're playing a vital role in database management.

Web developers use SQL to organize the website's database and marketers to help evaluate their marketing campaigns.

Rust

Programming Language Exercises

Multiple Choice.Choose the best answer for each question.

1.Which of the following is a primary purpose of a programming language?

(a) To confuse computer users
(b) To develop software programs and scripts for computers
(c) To design computer hardware
(d) To write essays

2.Which type of programming language is designed to be easier for humans to read and understand?

(a) Low-level language
(b) Machine language
(c) High-level language
(d) Assembly language

3.Which of the following is NOT typically considered a feature of a good programming language?

(a) Simplicity
(b) Obscurity
(c) Efficiency
(d) Abstraction

4.Which of the following programming languages is known for its use in web development to add dynamic elements to web pages?

(a) Java
(b) Python
(c) JavaScript
(d) C++

5.Which programming language is often associated with the .NET framework and is compatible with both Windows and Linux?

(a) Python (c) Java

(b) C# (d) PHP

6.Which language is particularly well-suited for data science, statistical computing, and managing relational databases?

(a) C++ (c) Go

(b) SQL (d) JavaScript

7.According to the text, which language is considered a good starting point for someone wanting to kickstart their career as a software developer or engineer?

(a) C++ (c) Python

(b) Java (d) Assembly Language

8.Which feature of the Go programming language makes it ideal for creating algorithms, web servers, and data pipelines?

(a) Simplicity (c) Object-oriented programming

(b) Concurrency (d) Platform independence

9.Which of the following companies is NOT mentioned as using Python?

(a) Intel (c) Spotify

(b) Facebook (d) Apple

10.Which of these languages is known for being particularly useful in finance, healthcare, and e-commerce?

(a) Python (b) JavaScript (c) Java (d)Go